Activity

TOOL: R Program **RegisterNo:192424335**

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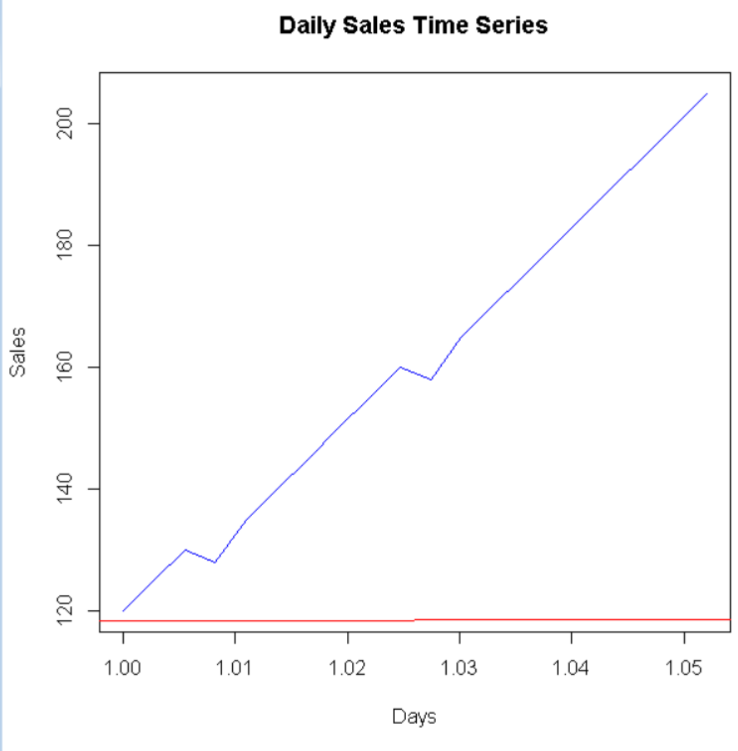
COURSE: Data Handling and Visualization **Code: DSA0604**

1. Daily Sales Time Series

Code:

```
sales <- c(120, 125, 130, 128, 135, 140, 145, 150, 155, 160,
         158, 165, 170, 175, 180, 185, 190, 195, 200, 205)
sales_ts <- ts(sales, frequency = 365)
plot(sales_ts,
    main = "Daily Sales Time Series",
    xlab = "Days",
    ylab = "Sales",
    col = "blue")
abline(reg = lm(sales ~ seq_along(sales)), col = "red")
```

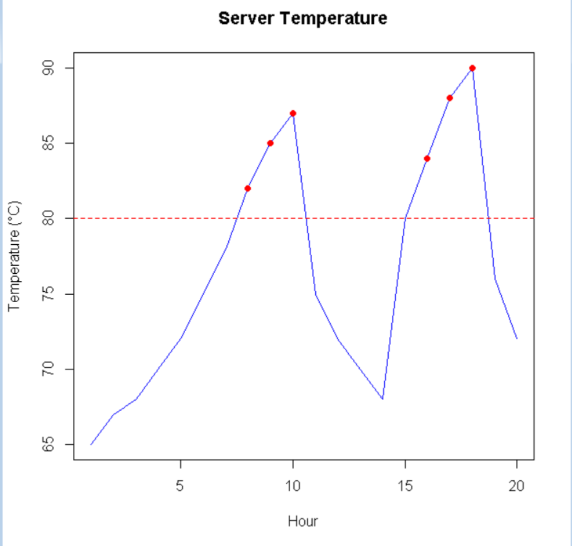


2. Server Temperature Monitoring

Code:

```
temperature <- c(65, 67, 68, 70, 72, 75, 78, 82, 85, 87,
```

```
               75, 72, 70, 68, 80, 84, 88, 90, 76, 72)
time <- 1:length(temperature)
safe_limit <- 80
plot(time, temperature,
     type = "l",
     main = "Server Temperature",
     xlab = "Hour",
     ylab = "Temperature (°C)",
     col = "blue")
abline(h = safe_limit, col = "red", lty = 2)
points(time[temperature > safe_limit],
       temperature[temperature > safe_limit],
       col = "red",
       pch = 19)
```

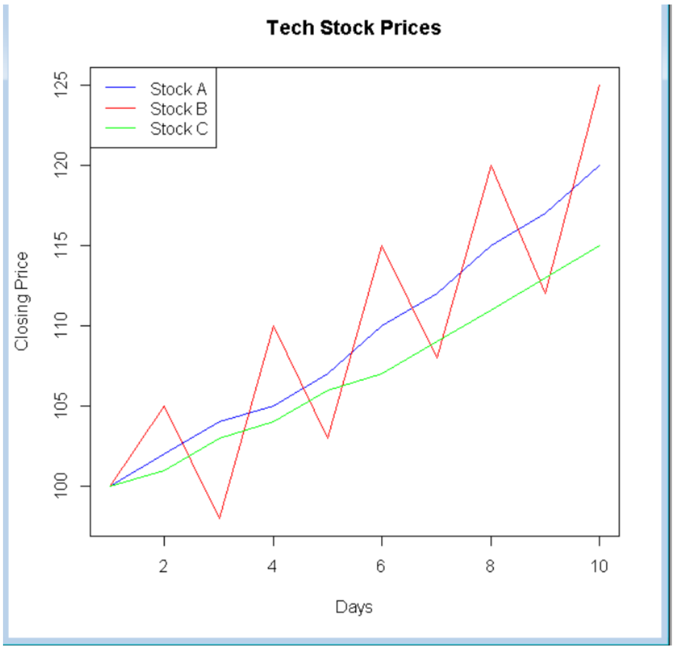


3. Tech Stock Comparison

Code:

```
stock_A <- c(100, 102, 104, 105, 107, 110, 112, 115, 117, 120)
stock_B <- c(100, 105, 98, 110, 103, 115, 108, 120, 112, 125)
stock_C <- c(100, 101, 103, 104, 106, 107, 109, 111, 113, 115)
plot(stock_A,
     type = "l",
```

```
    col = "blue",
    ylim = range(c(stock_A, stock_B, stock_C)),
    main = "Tech Stock Prices",
    xlab = "Days",
    ylab = "Closing Price")
lines(stock_B, col = "red")
lines(stock_C, col = "green")
legend("topleft",
       legend = c("Stock A", "Stock B", "Stock C"),
       col = c("blue", "red", "green"),
       lty = 1)
```

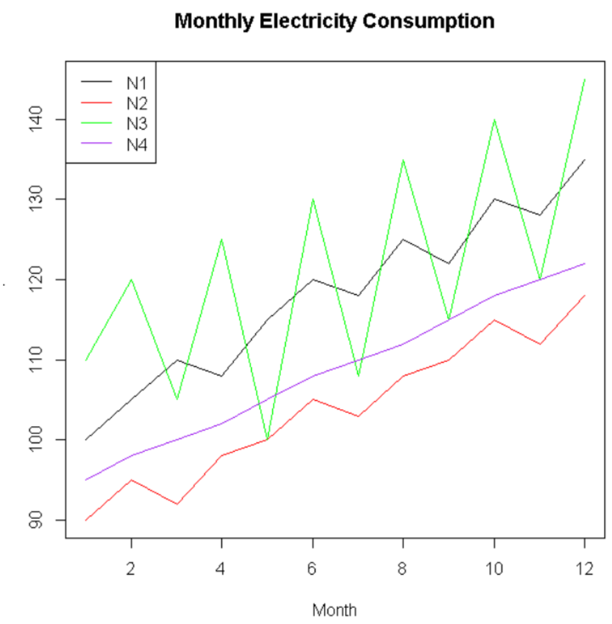


4. Neighborhood Electricity Consumption

Code:

```
month <- 1:12
N1 <- c(100, 105, 110, 108, 115, 120, 118, 125, 122, 130, 128, 135)
N2 <- c(90, 95, 92, 98, 100, 105, 103, 108, 110, 115, 112, 118)
N3 <- c(110, 120, 105, 125, 100, 130, 108, 135, 115, 140, 120, 145)
N4 <- c(95, 98, 100, 102, 105, 108, 110, 112, 115, 118, 120, 122)
plot(month, N1,
    type = "l",
    ylim = range(c(N1, N2, N3, N4)),
    main = "Monthly Electricity Consumption",
    xlab = "Month",
    ylab = "Consumption")
lines(month, N2, col = "red")
lines(month, N3, col = "green")
lines(month, N4, col = "purple")
legend("topleft",
      legend = c("N1", "N2", "N3", "N4"),
```

```
col = c("black", "red", "green", "purple"),
lty = 1)
```



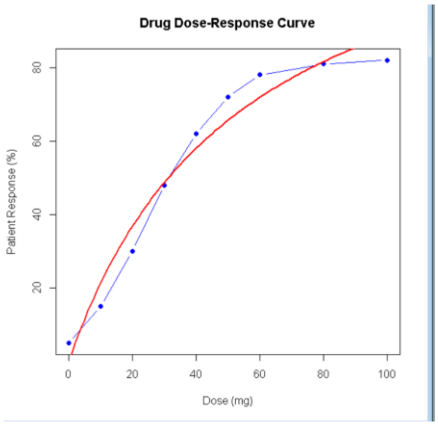
5. Drug Dose–Response Curve

Code:

```
dose <- c(0, 10, 20, 30, 40, 50, 60, 80, 100)

response <- c(5, 15, 30, 48, 62, 72, 78, 81, 82)
plot(dose, response,
     type = "b",
     pch = 19,
     main = "Drug Dose-Response Curve",
     xlab = "Dose (mg)",
     ylab = "Patient Response (%)",
     col = "blue")
model <- nls(response ~ a * dose / (b + dose),
             start = list(a = 100, b = 20))

curve(predict(model, data.frame(dose = x)),
      add = TRUE,
      col = "red",
```

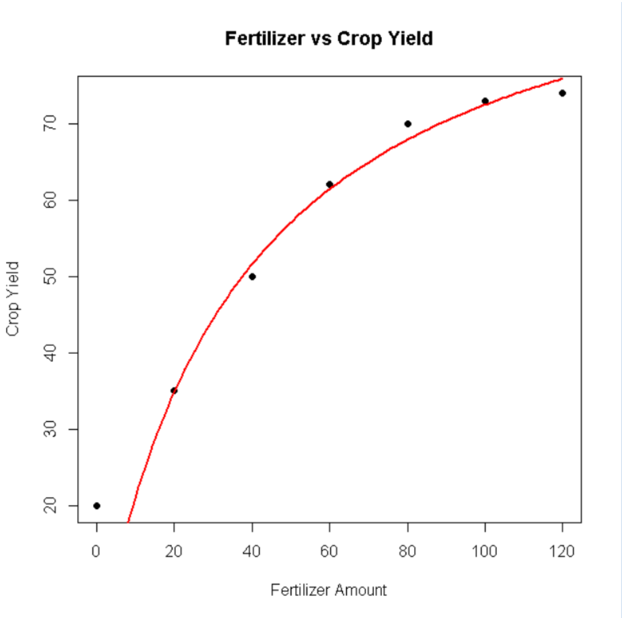


lwd = 2)

6. Fertilizer vs Crop Yield

Code:

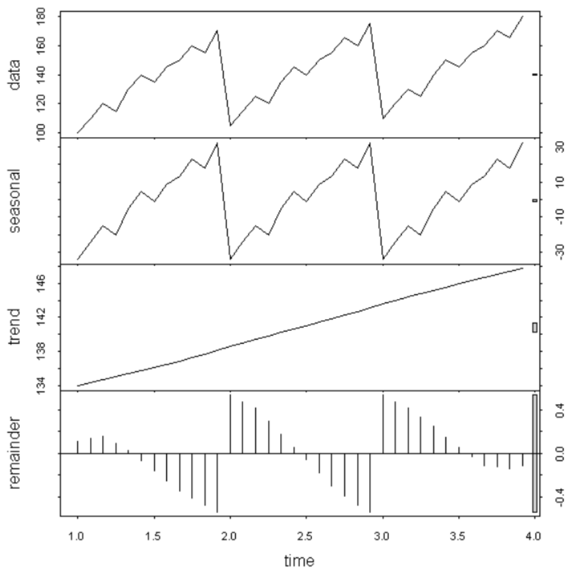
```
fertilizer <- c(0, 20, 40, 60, 80, 100, 120)
yield <- c(20, 35, 50, 62, 70, 73, 74)
plot(fertilizer, yield,
     pch = 19,
     main = "Fertilizer vs Crop Yield",
     xlab = "Fertilizer Amount",
     ylab = "Crop Yield")
model <- nls(yield ~ a * fertilizer / (b + fertilizer),
             start = list(a = 100, b = 30))
curve(predict(model, data.frame(fertilizer = x)),
      add = TRUE,
      col = "red",
      lwd = 2)
```



7. STL Decomposition of Ticket Sales

Code:

```
sales <- c(100, 110, 120, 115, 130, 140, 135, 145, 150, 160, 155, 170,
           105, 115, 125, 120, 135, 145, 140, 150, 155, 165, 160, 175,
           110, 120, 130, 125, 140, 150, 145, 155, 160, 170, 165, 180)
sales_ts <- ts(sales, frequency = 12)
decomp <- stl(sales_ts,
s.window = "periodic")
plot(decomp)
```



8. STL Decomposition of Weekly Users

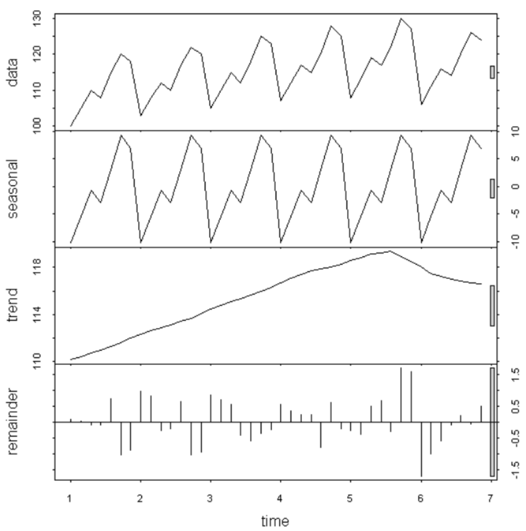
Code:

```
users <- c(100, 105, 110, 108, 115, 120, 118,
           103, 108, 112, 110, 117, 122, 120,
           105, 110, 115, 112, 118, 125, 123,
           107, 112, 117, 115, 120, 128, 125,
           108, 113, 119, 117, 122, 130, 127,
           106, 111, 116, 114, 120, 126, 124)

users_ts <- ts(users, frequency = 7)

decomp <- stl(users_ts, s.window = "periodic")

plot(decomp)
```

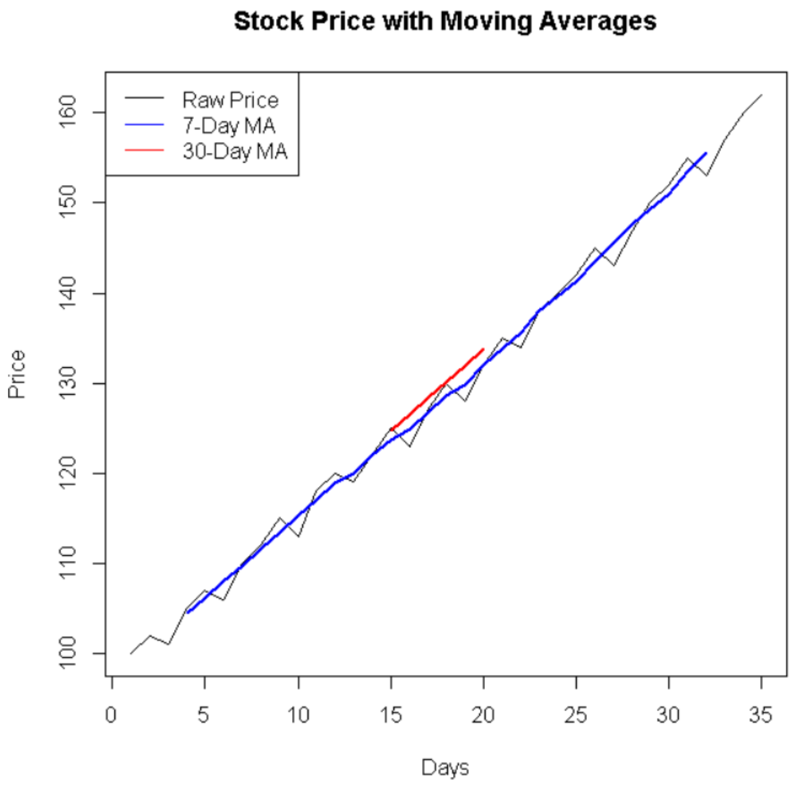


9. 7-Day and 30-Day Moving Average

Code:

```
price <- c(100,102,101,105,107,106,110,112,115,113,
        118,120,119,122,125,123,127,130,128,132,
        135,134,138,140,142,145,143,147,150,152,
        155,153,157,160,162)
MA7 <- filter(price, rep(1/7, 7), sides = 2)
MA30 <- filter(price, rep(1/30, 30), sides = 2)
plot(price,
    type = "l",
```

```
    main = "Stock Price with Moving Averages",
    xlab = "Days",
    ylab = "Price")
lines(MA7, col = "blue", lwd = 2)
lines(MA30, col = "red", lwd = 2)
legend("topleft",
       legend = c("Raw Price", "7-Day MA", "30-Day MA"),
       col = c("black", "blue", "red"),
       lty = 1)
```



10.GDP Detrending

Code:

```
gdp <- c(100, 103, 105, 108, 112, 115, 117, 121,
         125, 128, 130, 134, 138, 140, 143, 147)

time <- 1:length(gdp)
```

```
model <- lm(gdp ~ time)

trend <- predict(model)

detrended <- gdp - trend

plot(time, detrended,
    type = "l",
    main = "Detrended GDP",
    xlab = "Quarter",
    ylab = "Detrended GDP",
    col = "blue")

abline(h = 0, lty = 2)
```